FERITE-STEEL · CRM & BUSINESS MANAGEMENT SYSTEM

Module 4 — Credit Risk AI

Scope Definition & Decision Document

PREPARED FOR FERRITE STRUCTURAL STEELS PRIVATE LIMITED · RADHE ENTERPRISE

PURPOSE OF THIS DOCUMENT

This document defines what Module 4 will do and sets out the choices that need to be made before development begins. Sections 1–3 describe the module as agreed. Sections 4–6 present options on three open questions — **what the assessment outputs, how the customer is scored, and where the feature appears** — each with a recommendation. Section 7 lists information required from Ferrite Steel. Please mark your selections in the decision boxes and return this document.

1 What This Module Does

Before extending credit terms to a customer, your team currently relies on experience and judgement. Module 4 replaces this with a structured, AI-assisted assessment. A salesperson uploads the customer's financial documents, and the system produces a risk assessment and a clear recommendation — all in under a minute.

The assessment is a decision aid, not a decision maker. It gathers the evidence, applies a consistent standard to it, and puts a reasoned recommendation in front of the person who holds the authority. The final credit decision always rests with a lead or admin.

2 Step-by-Step Workflow

1. The salesperson opens a customer's profile in the CRM.
2. They click **"Run Credit Check."**
3. They upload the customer's financial documents — GST returns, bank statement, ITR; whichever are available.
4. They answer a short set of structured questions about the customer (see Section 5.2).
5. The system automatically extracts the content from the documents.
6. The AI analyses the documents, the questionnaire answers, and the customer's history with Ferrite Steel.
7. A **Credit Assessment Report** is generated and saved against the customer's profile.
8. The lead or admin reviews the report and makes the final credit decision.

3 What the AI Analyses

The assessment draws on three sources of information.

A · From your internal records

- Customer's business type — Commercial, Industrial, or Government.
- GST and PAN registration status.
- Past transaction history with Ferrite Steel — number of deals, total value, win/loss ratio, and payment-terms pattern.

B · From uploaded documents

- **GST returns** — revenue size and regularity of GST filing.
- **Bank statements** — cash flow and account balances.
- **ITR (Income Tax Return)** — declared income.
- **Historical financial documents** — where earlier years are available, the system compares periods to establish whether the business is growing, stable, or contracting. A single year shows position; multiple years show direction.

C · From structured questions

- A short questionnaire completed by the salesperson at the time of assessment, capturing knowledge that exists in your team's heads but not in any document — years of trading relationship, references from other suppliers, site or premises verification, and any known payment disputes.

> **WHY THE QUESTIONNAIRE MATTERS**
> Documents describe a business on paper. Your team's direct experience of a customer — whether they answer the phone when payment is due, whether their yard actually holds stock — is often the stronger signal. The questionnaire captures this in a consistent format so it can be weighed alongside the financial evidence rather than lost in conversation.

4 Decision One — What the Assessment Outputs

Every assessment produces a written summary and a list of key factors. The open question is what *form the headline verdict takes*. Three options are set out below; they differ in how much detail is presented and how much interpretation is left to the reader.

Option A — Risk Level Only

The assessment returns a risk level of **Low**, **Medium**, or **High**, a recommendation, and the written summary. No numeric score is shown.

+ Simplest to read; no numbers to argue about or misinterpret.
+ Cannot create false precision — a band honestly reflects a judgement.
− Cannot rank or sort customers within a band; every "Medium" looks alike.
− Harder to see improvement or decline in a customer over time.

Option B — Score with Risk Level RECOMMENDED

The assessment returns a **numeric score**, the risk level it falls into, a recommendation, and the written summary. The score allows comparison; the band drives the decision.

+ Customers can be ranked, sorted, and filtered — useful as the customer base grows.
+ Change over time is visible: a customer moving from 54 to 71 is a real signal.
+ The band still governs the action, so small score movements do not change outcomes.
− Requires discipline not to over-read small differences between scores.

Option C — Score, Risk Level and Suggested Credit Terms

As Option B, with the addition of a **suggested credit limit in rupees and a suggested credit period in days**, which the approver may accept or override.

+ Most complete output; answers not just "is this safe" but "how much, for how long."
+ Creates a formal record of approved limits that later modules can check against.
− Requires Ferrite Steel to define the rules for suggesting limits — typically a proportion of monthly turnover, capped by exposure appetite. This is a business policy decision, not a technical one.
− Adds development effort and a further approval step to the workflow.

DECISION 1 — OUTPUT FORMAT

☐ Option A — Risk level only
☐ Option B — Score with risk level *(recommended)*
☐ Option C — Score, risk level and suggested credit terms

4.1 Scale Options (if Option B or C is selected)

Scale	Character	Best suited to
1 – 10	Immediately intuitive; everyone reads it correctly without explanation.	Teams who want a quick verdict and will not analyse scores closely.
0 – 100 *(recommended)*	Finer resolution; distinguishes customers that a 1–10 scale would flatten together.	Ranking a growing customer base and tracking change over time.
Letter grades AAA – D	Mirrors the format of commercial credit-rating agencies.	Presenting assessments to banks or external parties.

DECISION 1B — SCALE

☐ 1 – 10 ☐ 0 – 100 *(recommended)* ☐ Letter grades

5 Decision Two — How the Customer Is Scored

This section covers two related questions: **what evidence carries what weight**, and **what mechanism produces the number**.

5.1 Proposed Weighting

Evidence is grouped into four pillars. Each is assessed separately, then combined. The weighting differs depending on whether the customer has traded with Ferrite Steel before, because past payment behaviour with you is the strongest available predictor of future payment behaviour.

Pillar	New customer	Existing customer
Trading history with Ferrite Steel	Not available	40%
GST filing & declared turnover	40%	25%
Bank statement health	35%	20%
ITR / declared income & questionnaire	25%	15%

These weightings are a starting proposal based on what typically predicts default in Indian B2B distribution. They can be adjusted before build, and revisited once the module has been in use.

5.2 What Each Pillar Looks For

Pillar	Signals assessed
Trading history	Length of relationship; number and value of completed deals; whether payments have been made on time; any dishonoured payments; whether agreed terms have been repeatedly stretched.
GST	Whether returns are filed every period without gaps; declared turnover and its trend; consistency between GST turnover and the figures shown in other documents.
Bank statements	Average closing balance; regularity of inflows; whether the account shows genuine trading activity; returned or bounced items; stability across the period.
ITR & questionnaire	Declared income and its consistency with GST and bank evidence; supplier references; premises verification; known disputes.

5.3 Automatic Warning Flags

Certain findings are serious enough that they should be raised prominently regardless of how the customer scores elsewhere. Proposed flags:

- A dishonoured payment in the trading history with Ferrite Steel.
- Gaps in GST filing over the assessed period.
- A material mismatch between GST turnover, ITR income and bank inflows.
- Returned or bounced items visible in the bank statement.

Ferrite Steel should decide whether a flag **caps the assessment at "High Risk" automatically**, or is **shown as a prominent warning** while the score is calculated normally.

DECISION 2A — TREATMENT OF WARNING FLAGS

- ☐ Flags automatically cap the assessment at High Risk
- ☐ Flags are displayed prominently but do not override the score *(recommended)*

5.4 Scoring Mechanism

Three approaches are available for producing the number itself.

Option 1 — Fixed Rules DETERMINISTIC

Thresholds are defined in advance and applied mechanically. For example: twelve consecutive GST filings scores full marks; each missing period deducts a set amount. The AI extracts figures from the documents but does not judge them.

- + Completely reproducible — identical inputs always produce an identical score.
- + Every point is traceable to a stated rule.
- − Rigid. A customer failing one threshold by a small margin is treated the same as one failing it badly.
- − Requires Ferrite Steel to define and defend every threshold in advance.

Option 2 — Guided AI Judgement RECOMMENDED

The AI assesses each pillar against a written rubric containing worked reference examples — a described profile and the score it should attract. The rubric guides the judgement without dictating it, allowing the AI to weigh context that fixed thresholds cannot capture.

- + Handles real-world documents, which are rarely uniform or complete.
- + Can recognise mitigating context — a single missed GST filing during a known disruption is not the same as chronic non-filing.
- + Rubrics can be refined over time as Ferrite Steel's experience of the module grows.
- − Two runs on the same documents may differ by a few points. See note below.

Option 3 — Unguided AI Judgement

The AI is given the documents and asked for an overall assessment without a defined rubric or weighting.

- + Fastest to build; no rubric definition required.
- − No consistent standard between customers — the basis of assessment may shift from case to case.
- − Difficult to explain or defend a specific outcome. **Not recommended for credit decisions.**

ON CONSISTENCY OF AI SCORING

Under Option 2, running the same documents twice may produce slightly different scores — for instance 68 on one occasion and 71 on another. In practice this does not affect decisions, for two reasons. First, **the risk band governs the action**, and both figures fall in the same band, producing the same recommendation. Second, **each assessment is saved permanently as a dated record**. Nobody re-runs an assessment to check a past decision; they open the stored report, which shows the score, summary and factors exactly as they stood on that date. Running a fresh assessment creates a new dated record alongside the old one, which is the intended behaviour when new documents become available. The score should be read as a considered judgement, not a measurement — differences of a few points carry no meaning.

DECISION 2B — SCORING MECHANISM

- ☐ Option 1 — Fixed rules
- ☐ Option 2 — Guided AI judgement *(recommended)*
- ☐ Option 3 — Unguided AI judgement

6 Decision Three — Where the Feature Appears

Credit assessment is most useful at the moment credit is actually being considered. The proposal is that assessments are **created** in one place and **viewed** in several, so the information reaches the person who needs it without them having to go looking.

Location	Function	Detail
Customer profile page	Create & view	The primary location. A "Run Credit Check" button, the customer's current credit status shown at a glance, and the full history of past assessments.
Credit Risk section (main navigation)	Create & view	A dedicated area listing all assessments across customers, filterable by risk level and date. Allows a new assessment to be started by selecting a customer.
Quotation editor	View only	The customer's current credit status displayed while a quotation is being prepared, so credit terms are set with the assessment visible. Read-only; no assessment is run from here.

> **WHY VISIBILITY INSIDE THE QUOTATION EDITOR MATTERS**
> A credit assessment that sits on a profile page only helps if somebody remembers to look at it. Showing the customer's current status at the point where credit terms are actually being written into a quotation is what turns the module from a record into a safeguard.

6.1 Optional Additions

The following can be included if wanted; each adds development effort.

- **Dashboard summary** — a breakdown of the customer base by risk level on the main dashboard.
- **Re-assessment reminders** — a prompt when a customer's most recent assessment passes a defined age, for example twelve months.
- **Notification on High Risk** — an alert to the admin whenever an assessment returns a High Risk result.

DECISION 3 — PLACEMENT

- ☐ Confirm the three core locations above
- ☐ Add dashboard summary
- ☐ Add re-assessment reminders Interval: _______ months
- ☐ Add High Risk notification to admin

7 Who Can Use It

Role	Access
Admin	Run assessments; view all results across all teams.
Team Lead	Run assessments; view results for their own team.
Member	No access.

8 Documents Required Per Customer

Collect the following from the customer before running a credit check.

- **GST Returns** — last 2–3 quarters (PDF).
- **Bank Statement** — last 6 months (PDF).
- **ITR** — last 1–2 years (PDF).
- **Earlier-year financials** — where available, to establish trend.

> **DATA HANDLING**
>
> Documents are processed at upload time and are **not stored on the server** — only the extracted text is retained for the AI to read. This keeps storage lean and avoids holding sensitive customer files long-term.

> **INCOMPLETE DOCUMENTS**
>
> If some documents are unavailable, the system still runs. The assessment will state what was and was not provided, and will indicate reduced confidence accordingly. An assessment based on partial evidence is marked as such so it is not mistaken for a complete one.

9 Information Required From Ferrite Steel

The following need to be confirmed before development begins. Items 1 and 2 are the most urgent, as they determine whether the trading-history pillar can be built at all.

#	Question
1	Where is customer payment history currently recorded — invoice dates, payment dates, and any dishonoured payments? Is this in SAP, in Tally, in spreadsheets, or held informally?
2	Can that payment data be exported, and in what format? Without it, the trading-history pillar cannot be scored and weightings must shift to the document pillars.
3	Are credit limits currently recorded anywhere formally, or are they decided case by case?
4	What credit periods does Ferrite Steel typically offer — 30, 45, 60, 90 days?
5	What is the largest credit exposure the business is willing to carry with a single customer?
6	Roughly how many customers are expected to be assessed per month? This determines running cost.
7	Are there past customers who defaulted, whose documents could be used to calibrate the rubric? Real examples of bad outcomes are the most valuable input available.
8	Should assessments be visible to the team lead who ran them only, or to all leads?
9	Which questions should the structured questionnaire ask? A draft will be proposed for review.

10 Module Summary

Module	4 — Credit Risk AI
Fee	₹20,000
Effort	60–75 hours
Inputs	Customer documents (PDF), structured questionnaire, internal transaction history
Output	Risk assessment, recommendation, written summary, key factors — final form per Decision 1
Dependencies	None — can start immediately after Phase 3 delivery
Prerequisite	Answers to Section 9, particularly items 1 and 2